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Department of Computer Science and Engineering



AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization

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Submitted by

Md. Rakib Hossain

Roll: 88 Registration No.: 2020315665

Shah Jamal Islam

Roll: 81 Registration No.: 2020915641

Supervised by

Dr. Saifuddin Md. Tareeq

Professor

Department of Computer Science and Engineering

University of Dhaka

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Declaration Form

This initial report presents the foundation of the thesis project entitled *AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization*. We declare that the background study, problem formulation and proposed methodology presented here are the work of the authors under the supervision of Dr. Saifuddin Md. Tareeq. Ideas, methods, datasets and findings originating from prior work are cited. The implementation and experimental results are intentionally reserved for later stages of the thesis. The present conclusion is limited to the background study and proposed methodology. This report is current to 7 September 2026; the project remains in progress until February 2027.

Md. Rakib Hossain
Candidate

Shah Jamal Islam
Candidate

Dr. Saifuddin Md. Tareeq
Supervisor

Abstract

Acknowledgements

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Chapter 1

Introduction

1.1 Context and Motivation

Business intelligence and analytics organize operational data so that decision makers can describe performance, anticipate future conditions and choose an appropriate response [4]. For an inventory-based small or medium enterprise (SME), this progression begins with familiar records: invoice lines, product movements, purchase receipts, expenses, payments and customer visits. These records can explain revenue and stock position, but their value is limited when they remain divided among paper ledgers, spreadsheets and separate applications. The owner may know that sales declined last month without knowing which products, customers or operating conditions caused the decline, whether the pattern is likely to continue, or which affordable action should be taken.

The decision is more demanding than producing a dashboard. Before Ramadan or Eid, for example, a retailer must estimate demand early enough to order from a supplier. Ordering too little creates stock-outs and forgone sales; ordering too much immobilizes working capital and may create obsolete stock. At the same time, a previously valuable customer may be becoming inactive, a product may appear profitable only because costs were recorded incompletely, or a promotion may increase volume without increasing contribution. These questions draw on the same operational history but require different analytical tasks and a common decision context.

Retail forecasting research identifies hierarchy, intermittency, promotions, new products, changing assortment and the organizational use of forecasts as central practical concerns [8]. Evidence from large forecast competitions also cautions against assuming that a more complex algorithm is automatically more accurate: combinations, statistical methods and carefully constructed baselines remain competitive across many series [17]. The relevant research problem is therefore not to attach a fashionable model to a sales file. It is to construct a defensible path from business events to out-of-time predictions and then from predictions to feasible, explainable actions.

1.2 Bangladesh SME Context

The intended setting is an inventory-holding retail SME in Bangladesh. This scope is narrower than the approved thesis title, but it makes the business entities, prediction units and decisions testable. Bangladesh-specific studies show that ICT adoption in SMEs is shaped by perceived benefit, management and government support, cost, complexity and the surrounding organizational and technological conditions [11, 22]. Research on mobile financial services further shows that users value accessibility, mobility, self-control and other affordances embedded in everyday practice [9]. These findings imply that local payment channels, low-friction data entry and owner-readable explanations are design requirements, not optional presentation details.

This study uses a functional definition of its population rather than relying on a regulatory size threshold that may change. A target business holds stock, records sales and purchases, makes branch- or product-level decisions, and has limited specialist analytics capacity. The unit of analysis may be a business, branch, stock-keeping unit (SKU), customer or historical customer snapshot, depending on the research question. Conclusions will be restricted to the participating firms and the evidence actually collected.

1.3 Decision Environment and Stakeholder Requirements

An SME analytics artifact is used inside an operating system of people, records, cash commitments and supplier relationships. The owner or manager is the principal decision maker, but the evidence may be created by sales staff, inventory staff, an accountant and external payment or supply services. A recommendation is dependable only when those roles produce compatible records and can understand the result at the point of action. The Bangladesh adoption evidence therefore has a direct methodological implication: perceived value, organizational readiness, cost and workflow compatibility must be evaluated alongside model accuracy [11, 22].

Table 1.1 translates this setting into concrete information requirements. The table is a design analysis for the present thesis, not a claim that every participating business has the same staffing structure. In a micro-enterprise one person may occupy several roles. The important distinction is between who creates an event, who validates it, who acts on an output and who bears the consequence of a poor decision.

TABLE 1.1: Stakeholder decisions and evidence requirements in the target setting.

Role	Decision or responsibility	Evidence required	Design implication
Owner or manager	Reorder, allocate cash, approve contact and accept or reject a recommendation	Current stock, expected demand, cost range, uncertainty and reason	Present a short feasible action with drill-down evidence and retain owner authority
Sales or inventory staff	Record sales, returns, receipts and adjustments	Stable product keys, timestamps, quantities and correction history	Minimize duplicate entry; validate at import and preserve corrections
Accounts function	Reconcile payments, expenses, receivables and margin inputs	Invoice-payment links, expense period and cost provenance	Separate cash movement, revenue and estimated profit; expose missing cost data
Supplier or logistics contact	Confirm lead time, pack size, minimum order and incoming quantity	Dated purchase orders and receipt history	Treat lead time and order restrictions as constraints, not fixed universal constants
Research and system team	Build, evaluate and monitor the artifact	Versioned datasets, cutoffs, code, parameters, metrics and audit events	Make every reported output reproducible from an immutable analytical snapshot

Four owner-facing qualities follow from this analysis. *Relevance* means that the output answers a decision the business can actually take. *Traceability* means that the displayed value can be followed back to source events, transformations and a declared cutoff. *Proportionality* means that model, interface and data requirements are reasonable for the decision's value and risk. *Contestability* means that an owner can inspect the supporting evidence, reject the suggestion and record a reason. These qualities turn local usability from a visual-design preference into a condition of credible evaluation.

1.4 Problem Statement

The central problem is the absence of a trustworthy and usable connection among three activities: recording SME operations, extracting predictive insight and selecting a feasible business action. A transaction system can preserve orders and stock but still provide no forecast. A standalone model can report an accuracy score but still be unusable because its features would not exist at the decision time, its test data leaked information from training, or its output is not linked to budget, stock, lead time and customer consent. A recommendation can be technically valid yet operationally impossible.

The problem is decomposed into four linked gaps:

- G1. **Data foundation gap.** Sales, inventory, purchases, expenses, customers and payments need a consistent schema, timestamps, provenance and organization boundaries before they can support comparable analytics.
- G2. **Prediction-validity gap.** Demand and customer-risk models must use historical snapshots, chronological or entity-safe splits, training-only transformations, strong baselines and metrics appropriate to sparse or imbalanced outcomes. Leakage can otherwise create persuasive but invalid performance [15].
- G3. **Decision gap.** A forecast or risk probability does not specify what to order, whom to contact or whether the action is affordable. Prediction must be connected to costs, constraints and uncertainty; this is the transition from predictive to prescriptive analytics [1].
- G4. **Adoption and accountability gap.** An SME owner needs a concise reason, confidence indication and traceable source for a suggestion. Feature attribution can describe model behavior, but it neither proves causality nor replaces human judgment [16, 18].

Accordingly, this thesis asks how an integrated, Bangladesh-aware framework can transform auditable retail records into leakage-controlled forecasts and customer insights, and then rank actions that respect the operating constraints of an SME. The work does not presume that machine learning will defeat every baseline. Discovering that a simple rule is more reliable or economical for a particular task is a valid outcome.

1.5 Analytical Logic and Claim Boundaries

The thesis separates four logical objects that are often collapsed in an analytics prototype: an observed business event, an analytical estimate, an eligible action and a later outcome. Figure 1.1 shows the dependency. Each transition has its own failure mode. Incomplete or inconsistent records weaken the evidence base; leakage or unsuitable validation invalidates an estimate; ignored constraints make an action infeasible; and missing outcome capture prevents learning whether the recommendation helped. Improvement at one stage cannot compensate for failure at another.

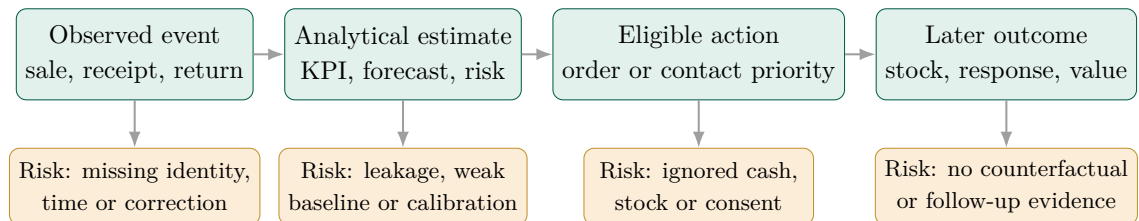


FIGURE 1.1: Logical dependency from an operational event to an evaluated outcome.

This separation also controls the claims made in the thesis. A dashboard value is *descriptive*: it is supported when the accounting definition and source reconciliation are correct. A forecast or inactivity score is *predictive*: it is supported only by performance on observations that were unavailable during model development. A ranked recommendation is *prescriptive*: it additionally requires explicit actions, costs, constraints and uncertainty [1]. A positive field impact claim is stronger still because it requires observed decisions and outcomes under a defensible comparison.

Several terms are therefore used narrowly. *Demand* denotes the quantity required by customers, whereas recorded sales may be a censored observation of demand when stock is unavailable. *Future inactivity* means no observed purchase during a predeclared horizon; it is not assumed to be a permanent or contractual churn state. *Explanation* means a faithful summary of model behavior and decision evidence; it is not proof that changing an attributed feature will cause a desired outcome [16, 18]. Finally, *local validity* means validity for the participating businesses under the sampled period and workflow, not automatic generalization to every SME in Bangladesh.

These boundaries make negative and conditional findings informative. A model may be accurate but not actionable because cost data are missing. An action policy may reduce simulated stock-outs only under a lead-time assumption that is too uncertain for deployment. An explanation may be technically faithful but misunderstood by owners. The evaluation is designed to reveal these conditions instead of merging them into a single optimistic success label.

1.6 Research Aim and Objectives

1.6.1 General Objective

The general objective is to design and evaluate an AI-powered business analytics framework that enables Bangladeshi retail SMEs to monitor performance, extract reliable and explainable insights, and prioritize feasible strategies.

1.6.2 Specific Objectives

The study will:

- O1. define an auditable canonical data model for sales, products, stock, purchases, returns, customers, suppliers, payments and expenses;
- O2. construct reproducible, time-indexed preprocessing and feature pipelines for performance analysis, SKU demand, customer inactivity and segmentation;
- O3. compare simple, statistical and machine-learning approaches under chronological or entity-safe validation, using both predictive and decision-oriented measures;

- O4. quantify uncertainty and provide global and case-level supporting factors without presenting association as causation;
- O5. formulate a constraint-aware mechanism for reorder, retention and operational action ranking; and
- O6. evaluate technical accuracy, robustness, scalability, usability and potential business value on clearly separated evidence tiers, culminating in consenting Bangladeshi SME data.

1.7 Initial Solution Concept

The proposed solution is B-SMART: the Bangladesh SME Monitoring, Analytics, Recommendation and Tracking framework. It is an end-to-end decision-support process rather than a new base classifier. Business events enter a governed operational layer; validated events are transformed into time-frozen analytical views; candidate models generate forecasts, risks and segment profiles; and a strategy layer removes infeasible actions before ranking the remainder. Each displayed action carries its data cutoff, model or baseline source, uncertainty and supporting factors. Owner response and later outcome are retained for evaluation and monitoring.

Three analytical outputs match the approved title. *Performance evaluation* summarizes revenue, margin, stock movement and customer behavior from reconciled records. *Insight extraction* estimates future demand and customer inactivity and identifies meaningful behavioral groups. *Strategy optimization* converts those estimates into candidates such as a reorder quantity or retention priority, subject to cash, storage, lead time, minimum order quantity, current stock and contact consent. Chapter 3 defines these components formally.

1.8 Research Questions

The thesis is guided by four questions:

RQ1.

Do calendar, price, promotion and market-context features improve out-of-time branch-SKU demand forecasts over seasonal and statistical baselines in the target SME setting?

RQ2.

How accurately and reliably can a historical customer snapshot identify no repeat purchase during a fixed future horizon?

RQ3.

Under the same demand and operating constraints, does a forecast-driven reorder policy reduce stock-out and holding cost relative to the participating business's current policy?

RQ4.

Do concise Bangla action explanations improve owner task completion, confidence and perceived usability without creating excessive decision time?

1.9 Scope and Boundaries

The research covers retail and inventory-based SMEs, structured operational data, descriptive indicators, demand forecasting, customer inactivity, behavioral segmentation, explanation and two strategy families: replenishment and consent-aware customer retention. The initial geographical context is Bangladesh; an external dataset may test method portability but cannot establish Bangladeshi business behavior. Public price data may supply market context but cannot substitute for invoice-level evidence.

The study excludes manufacturing scheduling, credit scoring, employee surveillance, fully autonomous price changes and unrestricted customer profiling. It does not claim causal effects from observational feature attributions. It will not treat generated data as field evidence, combine synthetic and real observations in a reported real-world score, or generalize from one retailer to all SMEs. Implementation detail, experimental results and final conclusions are deliberately reserved for Chapters 4–6.

1.10 Expected Contributions and Significance

The expected research contribution is the evaluated integration of established methods under constraints that matter to small retailers. Specifically, the work is expected to provide: (i) a traceable operational-to-analytical data design; (ii) reusable historical-snapshot protocols that prevent common forms of leakage; (iii) a Bangladesh-aware comparison of forecasting and customer analytics baselines; (iv) a transparent bridge from prediction to constrained action; and (v) evidence about whether owners can correctly interpret and use the resulting recommendations.

The practical significance is equally bounded. The proposed system is intended to help an owner see the evidence behind a decision, compare a recommendation with a familiar rule and retain authority over the final action. It is not intended to remove the owner from the decision or to label a statistically accurate output as profitable before business impact is measured.

1.11 Anticipated Challenges

The most consequential dependency is suitable primary data. The intended joint history of invoice lines, stock, products, prices and customers is difficult to obtain from a public Bangladesh source, and partner data may contain missing identifiers, changing SKU codes or incomplete cost records. Technical risks include intermittent demand, short histories, new products, stock-out censoring, class imbalance and drift. Scientific risks include target leakage, optimistic model selection, incompatible dataset comparisons and conclusions based on one firm. Human risks include time pressure, varied digital skill, low trust and over-trust in confident-looking explanations. The methodology assigns a specific prevention or evaluation step to each of these risks.

1.12 Organization of the Report

Chapter 2 develops the technical and contextual background, reviews the verified literature and identifies the research gap. Chapter 3 presents the research design, data protocol, formal tasks, proposed framework and evaluation plan. The Implementation and Experimental Results chapters remain named placeholders in this initial report. The Conclusions chapter synthesizes only what can be established from the present background study and proposed methodology; final outcome conclusions require the later empirical evidence.

Chapter 2

Background Study and Related Work

2.1 Background Study

2.1.1 Business Intelligence and the Decision Continuum

Business intelligence (BI) combines data, infrastructure, analytical methods and organizational practice to improve decisions. Chen, Chiang and Storey trace its development from structured enterprise reporting toward web, mobile and sensor-rich analytics [4]. This thesis adopts a three-stage decision continuum. *Descriptive analytics* establishes what happened and where; *predictive analytics* estimates what may happen next; and *prescriptive analytics* compares feasible responses. The stages are dependent: an action recommendation is only as credible as the prediction that supports it, and a prediction is only as credible as the operational data from which it was derived.

Operational and analytical workloads also have different integrity needs. A sale requires a consistent stock and payment update, whereas a model requires a historical feature view whose values do not change after evaluation. The proposed framework therefore separates the transaction record from versioned analytical snapshots. It retains the relationship between them through source identifiers, timestamps and transformation metadata.

2.1.2 SME Digitalization in Bangladesh

Technology adoption is not determined by model accuracy alone. A mixed-method case study of rural Bangladeshi SMEs identifies awareness of benefits, government support, top-management support and financial support as important ICT-adoption determinants [11]. A later survey of 535 SME managers combines the technology–organization–environment framework with diffusion-of-innovation constructs; relative advantage, complexity, perceived cost, management factors, competitive pressure and government support are among

the significant determinants reported [22]. These studies justify attention to cost, work-flow fit and organizational readiness, although neither evaluates a retail prediction system.

Mobile financial services are part of this operating context. Interview-based research in Bangladesh identifies accessibility, spatial and temporal mobility, self-control and related affordances in everyday MFS use [9]. Consequently, the proposed data model treats payment mode and settlement status as ordinary business fields. The research does not infer that a particular provider causes better performance; it asks whether locally recognizable records and explanations make the analytical workflow more usable.

2.1.3 Performance Evaluation and Data Foundations

Performance evaluation begins with well-defined measures. Revenue, gross margin, average order value, stock turnover, return rate and customer-repeat rate must each specify a time window, aggregation level and treatment of cancellations, refunds and missing costs. A dashboard value without these definitions can disagree with the accounting record or compare unlike periods. The canonical layer therefore preserves invoice lines and stock movements while deriving analytical measures in documented views.

Data provenance answers where a record came from, when it was received and how it was transformed. For primary SME data, provenance must be joined with organization isolation and pseudonymous customer identifiers. For public data, the license, original context and permissible inference must be recorded. A foreign transaction dataset can test whether code handles real invoices but cannot establish Bangladesh-specific customer behavior.

2.1.4 Data Quality, Provenance and Temporal Semantics

Data quality is not a single percentage. For this study it has at least five dimensions. *Syntactic validity* asks whether a value has the expected type, range and format. *Entity integrity* asks whether invoices, products, customers and branches have stable keys and valid relationships. *Temporal integrity* asks whether the event, correction, receipt and availability times are plausible and ordered. *Reconciliation* asks whether line amounts, payments, returns and stock movements agree under declared business rules. *Decision adequacy* asks whether the fields needed for a particular analysis are sufficiently complete at its decision cutoff. A file may score well on simple completeness while still be unusable for a margin analysis because unit cost is absent or for a customer study because identities are not stable.

This distinction matters because operational data are revised. A cancelled invoice may remain in an export, a return may arrive days after the sale, and a late stock adjustment may change the apparent availability history. Replacing the original value destroys the evidence needed to reconstruct what the system knew. The stronger design is append-aware: retain the source event, record the correction or status transition and build a

versioned analytical view for a declared cutoff. This implements the BI principle that analytical value depends on an intelligible relationship between data, infrastructure and organizational use [4].

Missingness must also be interpreted rather than automatically filled. Missing customer identity may mean an anonymous cash sale, not a failed import. A missing unit cost may make gross-margin analysis ineligible while leaving unit demand forecasting usable. A zero sale may represent zero demand, a closed shop, a missing file or an unobserved stock-out. These states have different meanings and should be separated whenever the source supports the distinction. Consequently, the proposed framework assigns eligibility at the level of a research task, business and period instead of declaring an entire dataset universally “clean” or “dirty.”

Provenance completes this quality model. Each derived field should identify its source columns, transformation version and cutoff; each reported result should identify the dataset manifest, split and model or rule version that produced it. This lineage does not guarantee correctness, but it makes errors auditable and results reproducible. It also prevents a public or generated dataset from quietly inheriting the claim status of primary Bangladesh SME evidence.

2.1.5 Customer Analytics: RFM, CLV and Segmentation

RFM summarizes *recency*, *frequency* and *monetary value* from a customer’s transaction history. These features are inexpensive to derive and retain a clear behavioral interpretation. Fader, Hardie and Lee connect RFM to customer lifetime value (CLV) through Pareto/NBD and gamma–gamma components and show how different histories can have similar expected value [7]. CLV is useful as an outcome or planning quantity, but a field used to define a segment cannot also be admitted as an independent predictor of that same segment.

Unsupervised segmentation asks whether coherent behavioral groups emerge without an existing label. For K-Means, the objective is

$$\underset{\{C_k, \mu_k\}_{k=1}^K}{\text{minimize}} \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2. \quad (2.1)$$

Because the procedure returns clusters even when the data form a continuum, cluster number and interpretation require validation. The silhouette coefficient compares within-cluster cohesion with separation from the nearest other cluster [20]. Stability under re-sampling and business usefulness are separate tests; an attractive label such as “loyal” does not establish that a cluster is statistically stable or actionable.

2.1.6 Retail Demand Forecasting

A demand forecast estimates the units required for a defined product, location and future horizon. Retail series may be affected by price, promotion, stock-outs, festivals, assortment changes and aggregation across products or branches. Fildes, Ma and Kolassa emphasize that retail forecasting practice cannot be reduced to a single algorithm because data hierarchy, operational process and forecast use matter alongside statistical accuracy [8].

Intermittent SKU demand contains many zero periods separated by nonzero demand. Croston’s method separately smooths nonzero size and the interval between demands [6]; Syntetos and Boylan analyze bias and accuracy in intermittent-demand estimates [24]. These works motivate an intermittent-demand baseline instead of forcing every SKU through a dense-series model. At the same time, forecasts must reconcile across product, category, branch and business totals. Optimal-combination reconciliation offers a principled way to produce coherent hierarchical forecasts [12].

Model complexity should follow the evidence. ARIMA-family models represent serial dependence after transformation; tree ensembles can learn nonlinear effects from lagged, calendar and price variables; and sequence models use ordered histories. Random Forest and XGBoost provide strong tabular comparators [2, 5]. The M5 retail competition demonstrates the importance of large-scale, hierarchical and uncertainty-aware comparison [17]. A seasonal-naive forecast is therefore a required benchmark, not a ceremonial baseline.

Forecast accuracy and inventory usefulness are related but not identical. Hyndman and Koehler demonstrate defects in common percentage errors and propose MASE for scale-free comparison [13]. Syntetos and colleagues connect forecast selection with stock-control performance in a wholesaling study [23]. The methodology accordingly reports point error, uncertainty and simulated stock-out/holding cost rather than selecting a method from one convenient metric.

2.1.7 Forecastability, Demand Regimes and Decision Loss

The forecasting task changes with the information content of a series. A high-volume staple with repeated weekly behavior offers many observations of level and seasonality. A slow-moving spare item may contain mostly zero periods, while a newly introduced product has little or no direct history. Pooling these cases under one algorithm and one average score can hide systematic failure. The literature on intermittent demand therefore supports a regime-aware comparison in which zero frequency, nonzero-demand variability and available history influence method eligibility [6, 24].

Temporal aggregation creates a further trade-off. Daily data preserve the timing needed for short-horizon ordering but may be too sparse for stable model estimation. Weekly aggregation reduces sparsity but may conceal festival peaks, stock-out intervals and short

lead-time decisions. Product aggregation provides more signal, yet a category forecast cannot directly determine which SKU to reorder. Hierarchical reconciliation addresses coherence across aggregation levels [12]; it does not remove the need to select a granularity that matches the operational decision.

Observed sales are also an imperfect target when availability is constrained. If an item is out of stock, a zero or low recorded sale does not establish that customers wanted zero units. Training on such periods without a stock indicator can teach a model that constrained supply is low demand. The appropriate response depends on evidence: flag and stratify censored periods when stock history exists, conduct sensitivity analysis when it is incomplete, and avoid claiming unconstrained demand when availability cannot be reconstructed.

Forecast errors have asymmetric decision consequences. Under-forecasting can produce shortage or lost-sales cost; over-forecasting can create holding, expiry or cash-opportunity cost. The best model by MAE need not minimize these consequences, particularly when errors occur on different products or at different times. This is why scale-aware error measures, prediction intervals and identical-demand-path inventory simulation serve complementary purposes [13, 23]. Model comparison remains prediction-focused until cost assumptions are declared; only then can a decision-oriented comparison be defended.

The review consequently rejects three shortcuts: choosing a deep model before measuring series structure, selecting from a single percentage error, and treating a forecast competition result as proof of value in a small local retailer. M5 provides strong evidence about comparative retail forecasting at scale [17]; the present task is to test which parts transfer under substantially different data volume, infrastructure and decision cost.

2.1.8 Customer Inactivity, Class Imbalance and Calibration

In a subscription service, churn may be an observed cancellation. In non-contractual retail it must be defined at a cutoff t : no repeat purchase during a fixed interval $(t, t+H]$. Every feature must be known by t , and the future label window must not contaminate training or validation. This definition turns an ambiguous business word into a reproducible prediction task.

The positive class may be rare or change across businesses. Saito and Rehmsmeier explain why precision–recall analysis is more revealing than ROC analysis for imbalanced binary tasks [21]. Model comparison must therefore report prevalence and ranking quality before action cost and benefit are considered in the prescriptive layer.

Ranking is not enough when the output is used in a decision calculation. A probability of 0.7 should correspond, approximately, to the event frequency among comparable cases. Niculescu-Mizil and Caruana compare probability estimates and post-hoc calibration methods for supervised learners [19]. The proposed framework therefore reports Brier score and calibration in addition to PR-AUC, ROC-AUC and top-fraction lift.

2.1.9 From Customer Score to Responsible Retention Decision

A customer-risk model creates a ranking or probability, not permission to act. For a fixed contact capacity, ranking quality determines whether the selected fraction contains more future inactive customers than an untargeted policy. For an expected-value calculation, calibration becomes essential because the probability is multiplied by uncertain benefit and cost. The operational threshold should therefore be selected from validation data using the declared contact capacity or utility assumptions, never by inspecting the final test.

The customer record creates additional ambiguity. Frequency and monetary value can reflect opportunity to visit, household purchasing, shared identifiers, returns or discount-seeking behavior; none alone proves loyalty or future value. Fader, Hardie and Lee show why the joint transaction history carries more information than a single RFM label [7]. Segmentation can help describe patterns, but assigning a persuasive segment name is an interpretive step that requires stability checks and owner review, not merely a clustering output [20].

Responsible use adds an independent feasibility gate. A high inactivity score does not justify contact when consent, contact policy or an appropriate offer is absent. Nor should an attribution such as “long recency increased risk” be restated as a causal claim that a specific message will retain the customer. The thesis therefore evaluates inactivity prediction, probability reliability, contact eligibility and later response as separate links. This structure makes it possible for the model to be scientifically informative even when the available evidence is insufficient for a profit or retention-effect claim.

2.1.10 Leakage, Explainability and Change Over Time

Data leakage arises when training has access to information that would not legitimately exist at prediction time. It may enter through a target-derived field, a post-outcome variable, repeated customers in both sides of a split or a transform fitted before the split [15]. Leakage control is therefore a property of the entire data pipeline, not a model option. A clear prediction time, feature availability table and split policy precede training.

SHAP represents a prediction as additive feature contributions relative to a reference value and supports local and global inspection [16]. Such an attribution describes how a fitted model used encoded features. Miller’s synthesis of social science research emphasizes that useful explanations are selective, contextual and often contrastive [18]. Thus a recommendation card should answer “why this action now?” in a small number of decision-relevant factors, while preserving access to fuller evidence and avoiding causal wording.

Business conditions also change. Retail monitoring must distinguish data-quality failure, input-distribution shift and outcome-performance decay. A retraining schedule alone is not evidence that a model remains valid; every replacement model must repeat the declared validation protocol.

2.1.11 From Prediction to Constrained Action

Prescriptive analytics uses predictions within an explicit decision problem. Bertsimas and Kallus formulate data-driven prescriptions in which auxiliary observations inform conditional stochastic optimization [1]. The present thesis adopts the underlying principle—evaluate actions by expected benefit, cost and constraints—without claiming to reproduce that paper’s asymptotic method. Reorder and retention are kept as separate action families because their outcomes, costs, permissions and uncertainties differ.

The complete framework is an information-system artifact as well as an analytical study. Design-science research builds and evaluates artifacts in relation to a relevant problem and a knowledge base [10]. This supports an evaluation plan that includes predictive validity, software behavior, owner usability and decision consequences instead of reporting model accuracy as the sole measure of success.

2.1.12 Cross-Stream Evidence Synthesis

No single literature stream supplies the complete argument for the proposed artifact. Bangladesh adoption studies establish contextual requirements but do not test transaction-level models. Forecasting and customer analytics provide task methods but usually begin after an analytical dataset has already been constructed. Leakage, calibration and explanation research provide safeguards but not an SME operating workflow. Prescriptive analytics connects estimates to decisions, while design science requires that the resulting artifact be evaluated against its intended problem. Figure 2.1 shows how these bodies of evidence contribute different, non-substitutable requirements.

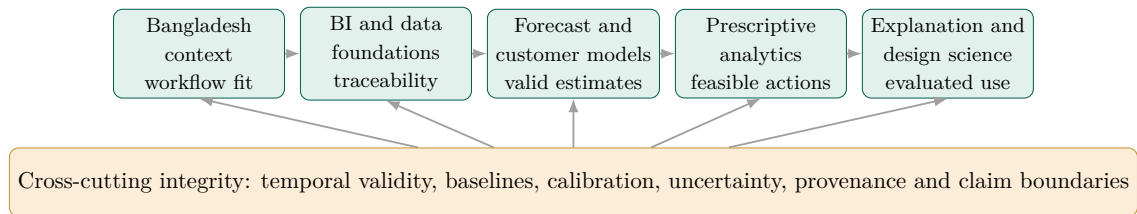


FIGURE 2.1: Complementary roles of the literature streams in the proposed design.

The arrows do not imply that the process is purely linear. Local workflow can change which data fields are available; data adequacy constrains model choice; uncertainty can remove an otherwise attractive action; and owner evaluation can reveal that an explanation or recommendation is unusable. The figure instead expresses logical dependence: later claims require the earlier evidence to be credible, while the cross-cutting integrity conditions apply to every stage.

This synthesis leads to a stricter definition of integration. Integration is not the co-location of a dashboard, a forecast and a classifier in one user interface. It requires a

shared operational vocabulary, compatible decision cutoffs, explicit task eligibility, separate uncertainty statements and a common audit trail from input to outcome. These properties become evaluation targets in Chapter 3.

2.2 Literature-Review Procedure

The present review is a structured background study, not a claim of an exhaustive systematic review. Sources were selected for direct relevance to one of six themes: Bangladesh SME digitalization, BI and design science, customer analytics, retail demand/inventory, churn and evaluation integrity, and explainable or prescriptive analytics. Searches used combinations of terms such as *Bangladesh SME ICT adoption*, *retail demand forecasting*, *intermittent demand*, *RFM customer segmentation*, *churn profit*, *data leakage*, *probability calibration*, *concept drift* and *prescriptive analytics*.

Peer-reviewed journal or conference work was preferred. A source was retained when it defined a required concept, supplied a method or evaluation principle, or provided Bangladesh-specific evidence. Metadata for every DOI-bearing entry was checked against Crossref, and dataset records were checked against DataCite or the official repository. Specific numerical claims were included only when supported by the paper metadata, abstract or full record available to the authors. The core set contains 22 scholarly works; three candidate datasets are listed separately in Chapter 3, giving exactly 25 cited references in this initial report.

2.3 Critical Comparative Analysis

2.3.1 Unit of Evidence and Temporal Boundary

The reviewed studies operate at different units of evidence, and their findings cannot be combined without respecting those units. An adoption survey may explain variation in managers' reported intention or organizational readiness; it does not supply repeated SKU outcomes. A retail forecasting benchmark treats each series and origin as a prediction case; it does not observe whether a small business could act on the result. A customer model treats a historical customer snapshot as a case; it cannot use a later purchase to describe that snapshot. Design-science evaluation, in turn, treats the artifact in context as the object of study [10]. These are complementary forms of evidence, not interchangeable sample sizes.

Time is the principal boundary connecting them. For a decision at cutoff t , an operational event is admissible only if it was recorded and available by t ; a feature summarizes only admissible events; an outcome occurs after t ; and any later correction must be versioned rather than inserted silently into the original snapshot. This rule is stronger than sorting a completed dataset by date. It requires the researcher to distinguish event time, recording

time and analytical availability. The leakage taxonomy described by Kaufman et al. shows why apparently innocuous variables or preprocessing choices can violate this separation [15].

Table 2.1 states the principal units and the strongest claim each can support. It is intentionally conservative. For example, accurate historical forecasts do not establish inventory benefit until a policy is tested on the same demand path, and a simulated policy benefit does not establish that an owner will use the recommendation correctly.

TABLE 2.1: Analytical units, temporal boundaries and defensible claims.

Analytical object	Unit and boundary	Defensible evidence	Invalid inferential leap
KPI	Business, branch or product within a closed reporting window	Reconciled numerator, denominator and data-quality status	Treating an unreconciled dashboard value as accounting truth
Demand forecast	Branch-SKU origin with future horizon hidden	Out-of-time error and uncertainty by demand regime	Inferring stock benefit from lower average error alone
Inactivity score	Customer snapshot at cutoff with a future label window	Ranking, calibration and lift on untouched snapshots	Treating inactivity as permanent churn or response to contact
Segment	Customer feature vector at a declared cutoff	Cohesion, separation, stability and domain interpretation	Assuming algorithmic clusters are natural or causal customer types
Reorder policy	Product decision on an identical replayed demand path	Stock-out, inventory and cost differences under stated assumptions	Claiming realized profit without field outcomes
Explanation	Owner decision task paired with model evidence	Correctness, time, confidence and qualitative interpretation	Inferring understanding from the presence of a SHAP chart

2.3.2 Comparator Adequacy and Model Eligibility

The comparative literature does not support a universal model hierarchy. Random Forest and XGBoost are strong general-purpose tabular learners [2, 5], but their suitability depends on feature availability, sample size, drift and the loss attached to error. Intermittent demand requires comparators designed for sparse occurrence patterns [6, 24]. Large retail competitions demonstrate the continuing importance of statistical structure, ensembles and well-constructed baselines [17]. A credible experiment must therefore ask whether a candidate is eligible before asking whether it wins.

Eligibility includes at least four tests. First, the training history must contain enough independent temporal variation for the candidate’s complexity. Second, all required features

must be available at the intended prediction time. Third, the candidate must produce the output needed by the decision, such as a calibrated probability or interval rather than only a point ranking. Fourth, its operational cost and failure behavior must be proportionate to any measured improvement. A marginal accuracy gain accompanied by unstable probabilities, opaque failure or infeasible latency is not automatically the best system choice.

Baseline selection is part of the research hypothesis. Seasonal-naïve and moving-average forecasts test whether learned calendar and context effects add value beyond recent repetition. Croston-family methods test whether sparse demand benefits from separate size and interval estimation. A recency rule and base-rate customer model test whether a classifier adds information beyond the strongest obvious behavior. Current-owner or reorder-point policies provide the operational comparator for a forecast-driven action. Without these comparators, a model can appear successful merely because the task is predictable at all.

2.3.3 Metric Alignment and Decision Consequences

Each metric answers a limited question. MAE summarizes absolute point error; MASE supports scale-aware comparison; WAPE emphasizes total absolute deviation relative to observed volume; interval coverage and width describe uncertainty. None alone measures cash tied in inventory. Similarly, ROC-AUC measures ranking across thresholds, whereas precision–recall analysis is often more informative about retrieving a minority outcome [21]. The Brier score and reliability analysis ask whether predicted probabilities are numerically trustworthy [19]. Silhouette values address geometric cluster separation, not commercial usefulness [20].

The evaluation must preserve these distinctions before joining them. A forecast candidate first needs valid out-of-time predictive evidence. A replenishment policy then converts the forecast distribution into an order under lead time, cash, storage and pack-size constraints. Only the second comparison can estimate holding and shortage consequences, and those estimates remain conditional on the cost model. The same logic applies to customer analytics: ranking and calibration precede contact prioritization; contact outcome and incremental value require additional field evidence.

This layered evaluation guards against metric substitution. It prevents high ROC-AUC from being reported as a profitable campaign, low WAPE from being reported as reduced stock-outs, and an internally coherent cluster from being reported as a stable market segment. It also permits a useful qualified result: a model may improve one analytical property without improving the subsequent decision under the current constraints.

2.3.4 Contextual Transfer and Evidence Boundaries

External evidence is necessary for method development but insufficient for a local effectiveness claim. UCI retail invoices can expose realistic returns, quantities and transaction

irregularities; a public Bangladesh demand series can exercise a local quantity-forecast procedure; market-price data can provide external context. Their schemas, units, customers and data-generating processes are different. Pooling their outcomes into one accuracy number would obscure, not strengthen, validity.

Bangladesh-specific studies reduce a different uncertainty. They show that cost, management support, relative advantage, workflow and everyday digital affordances matter in the local setting [9, 11, 22]. They do not establish the predictive effect of a festival feature or a reorder policy. The synthesis is therefore asymmetric: contextual studies inform requirements and evaluation questions; primary transaction records test local analytical claims; external datasets test portability or software behavior within an explicit boundary.

Generalization must consequently be decomposed. *Temporal generalization* concerns later periods in a participating firm. *Entity generalization* concerns unseen customers, products or branches. *Organizational generalization* concerns another business, and *geographical generalization* concerns a wider Bangladesh population. Success at an earlier level does not establish the next. The planned reporting will state which level each split and result actually evaluates.

2.3.5 Integration Cost and Failure Propagation

An integrated system creates value by sharing governed operational evidence, but integration also propagates failure. An incorrect product mapping can distort a KPI, a lag feature, an inactivity history and an order proposal at the same time. A late return can affect sales, customer value and inventory. An unversioned correction can make a previously reported model score impossible to reproduce. These dependencies explain why the central research gap is not closed by assembling independently accurate models behind one dashboard.

The appropriate response is controlled coupling. Tasks share canonical events, identities and provenance, but each analytical view has its own eligibility rules, cutoff, target and evaluation. A failure in unit cost can block a margin or profit claim without blocking unit-demand forecasting. Missing customer identity can block inactivity analysis without invalidating branch-level sales. This task-level isolation is especially important in SMEs, where demanding a perfect enterprise dataset before any analysis would make the artifact impractical.

The review therefore yields six design principles: preserve source evidence; define decision time before features; use task-specific eligibility; compare against a meaningful baseline; separate predictive, decision and user claims; and retain an accountable outcome loop. These principles are not additional algorithms. They are the logic that makes the later algorithms testable and the resulting recommendations contestable.

2.4 Thematic Synthesis of Prior Work

2.4.1 Business and Bangladesh Evidence

The BI literature supplies the descriptive–predictive–prescriptive frame [4], while design science supplies an artifact-oriented research logic [10]. Bangladesh studies explain why organizational support, perceived value, cost, complexity and actual local use should influence the proposed design [9, 11, 22]. Their limitation for the present problem is not quality but unit of analysis: adoption surveys and interviews do not provide invoice-level evidence for demand, inactivity or inventory decisions.

2.4.2 Customer Evidence

Fader et al. provide a probabilistic link between transaction history and future value [7], while the silhouette method provides a separate check on whether candidate behavioral clusters are internally coherent [20]. Together they motivate behavior-based profiles, but they do not by themselves establish a leakage-safe, non-contractual inactivity model for Bangladeshi retail.

2.4.3 Forecasting and Inventory Evidence

M5 supplies large, retail-specific comparative evidence [17]. Retail review evidence [8], intermittent-demand methods [6, 24], hierarchical reconciliation [12] and stock-control evaluation [23] show why an SME comparison requires more than an LSTM and one error column.

2.4.4 Risk, Integrity and Action Evidence

Precision–recall analysis and probability calibration address two different properties of a risk output [19, 21]. Leakage research explains why apparently excellent results can be unusable [15]. SHAP and human-centred explanation research then address inspection and communication [16, 18]. Finally, prescriptive analytics motivates placing predictions inside a cost-and-constraint model [1]. The unresolved question is whether these principles can be joined into a low-resource SME workflow and evaluated on the same temporal evidence chain.

2.5 Literature Matrix

Table 2.2 condenses the core set. “Relevant boundary” identifies what remains outside a source’s own objective; it is not a criticism of that work.

TABLE 2.2: Literature matrix for the 22-work core scholarly set.

Ref.	Context or data	Method	Main contribution used here	Relevant boundary for this thesis
[4]	Cross-domain BI research	Framework and bibliometric review	Descriptive-to-predictive BI evolution	No local SME artifact evaluation
[10]	Information-systems research	Design-science framework	Build-and-evaluate research logic	Does not define the retail analytics solution
[11]	Rural Bangladesh SMEs	Case study and questionnaire analysis	Local ICT-adoption determinants	No transaction-level prediction task
[22]	535 Bangladesh SME managers	TOE/DOI model with PLS-SEM	Organizational and technology determinants	Adoption intent, not business impact
[9]	Bangladesh MFS users	Interviews and thematic analysis	Everyday MFS affordances	User context rather than retail forecasting
[7]	CDNOW customer cohort	Pareto/NBD and gamma-gamma	Formal RFM-CLV relationship	No stock or action constraints
[20]	General clustering problems	Silhouette coefficient	Internal cluster validation	Does not establish business usefulness
[2]	General prediction benchmarks	Random Forest	Robust tree ensemble	Probabilities/actions require further checks
[5]	Large sparse benchmark tasks	Regularized tree boosting	Scalable nonlinear tabular learning	Does not prevent leakage by itself
[13]	Forecast evaluation examples	Accuracy-measure analysis	MASE and metric cautions	No inventory or owner outcome
[17]	Hierarchical Walmart retail series	Retail forecast competition design	Hierarchy, scale and uncertainty setting	Resource level unlike small shops
[8]	Retail research and practice	Critical review	Retail-specific data/process agenda	Does not implement the proposed local system
[6]	Intermittent stock demand	Separate size/interval smoothing	Foundational intermittent-demand method	Requires modern comparative evaluation
[24]	Intermittent-demand estimates	Bias and accuracy analysis	Corrected evaluation of estimators	No Bangladesh retail test
[12]	Hierarchical time series	Forecast reconciliation	Coherent forecasts across aggregation levels	Does not select business actions
[23]	Wholesaling demand and inventory	Forecasting/stock-control study	Links forecast choice to stock performance	Different organization and cost structure
[21]	Imbalanced binary problems	ROC/PR analysis	Value of precision-recall evaluation	Not a retail modelling study
[19]	Supervised benchmark datasets	Probability calibration comparison	Reliable probabilities as separate objective	Temporal drift remains outside evaluation
[15]	Competition and industry cases	Leakage formulation and taxonomy	Detection and prevention principles	Domain-specific availability must be mapped
[16]	Predictive models	Shapley additive explanations	Unified local feature attribution	Attribution is not causal evidence
[18]	Social science and AI literature	Interdisciplinary review	Selective, contrastive explanation principles	Requires owner-centred validation
[1]	Decision problems	Prescriptive optimization	Prediction-to-action bridge	Simplified here for SME constraints

2.6 Limitations of Existing Approaches

TABLE 2.3: Recurring limitations and the proposed methodological response.

Recurring limitation	Consequence	Response proposed in B-SMART
Static transaction re-reporting	Describes a problem after it occurs	Time-indexed forecasts and risk estimates tied to operational records
Model-only study	Score is disconnected from stock, cash and workflow	Constrained action generation and outcome tracking
Random temporal split	Future or repeated-entity information inflates results	Historical cutoffs, grouped splits and purged outcome windows
Accuracy-only selection	Good average error may produce poor inventory/customer decisions	Calibration, lift, uncertainty and business-cost evaluation
One model for every series	Intermittent, short and dense demand are treated alike	Baseline ladder and data-volume/series-type eligibility rules
Incoherent aggregation	SKU forecasts disagree with category or branch totals	Hierarchical reconciliation when the data support it
Black-box output	Owner cannot judge why action is suggested	Selective supporting factors, source and confidence statement
Foreign or generated evidence	Local validity is overstated	Explicit evidence tiers and claim restrictions
Unmonitored deployment	Performance decay and changed inputs remain hidden	Data-quality, drift, calibration and outcome monitoring

2.7 Research Gap and Thesis Positioning

The gap is derived by comparing what each literature stream establishes with what the stated SME decision problem requires. It is deliberately bounded to the 22-work scholarly core set; the thesis does not claim that no related system exists anywhere. Six connected gaps emerge.

G1—Contextual and empirical gap. Bangladesh research explains determinants and lived affordances of digital adoption [9, 11, 22], but it does not provide a joint, longitudinal test of SKU demand, customer inactivity and constrained retail actions. Conversely, large retail and forecast studies offer strong analytical evidence [8, 17], but their scale, market and infrastructure do not establish validity for owner-managed Bangladeshi SMEs. Local adoption evidence and transaction-level analytical evidence therefore remain disconnected.

G2—Operational-to-analytical integration gap. BI and CRM research show how operational data can support multiple analytical functions [4]. The forecasting, segmentation and churn studies in the matrix generally begin from an already prepared analytical dataset. They do not jointly evaluate the path from an auditable business event, through a reproducible historical snapshot, to the prediction and later action outcome. For the target SME, this missing path matters because a model cannot be reproduced if invoice corrections, stock movements, organization scope and feature cutoffs are not traceable.

G3—Methodological-validity gap. Prior work separately establishes the risks of leakage [15], intermittent demand [6, 24], incoherent hierarchical forecasts [12], unsuitable error measures [13] and uncalibrated classification probabilities [19]. What is missing is a single SME evaluation protocol that applies these safeguards together: decision-time feature availability, temporal/entity-safe splits, baseline eligibility by series type, coherent aggregation, calibration and an untouched final test.

G4—Prediction-to-decision gap. Accurate forecasts or customer rankings are intermediate outputs. Stock-control and profit-aware churn studies demonstrate that operational consequences can change model choice [23]; prescriptive analytics formalizes the use of predictions inside a decision problem [1]. The unresolved need is a transparent, low-resource mechanism that converts the outputs of several SME analytical tasks into actions while checking cash, stock, lead time, order rules, contact consent, uncertainty and action cost.

G5—Owner-centred explanation gap. SHAP provides principled feature attributions for complex predictions [16], while human-explanation research shows that useful explanations are selective and contextual [18]. Neither technical attribution alone nor a generic dashboard establishes that a Bangladeshi SME owner understands the evidence, distinguishes association from cause, and completes the intended decision correctly. Explanation quality must therefore be evaluated as owner task performance, not inferred from the presence of an explanation chart.

G6—Lifecycle-accountability gap. Business relationships between inputs and outcomes may change, and design science requires evaluation of the artifact in relation to its problem [10]. The reviewed task studies do not collectively close the local loop by recording model/baseline version, data cutoff, recommendation, owner response, later outcome and revalidation decision. Without that record, a system cannot show whether a recommendation remained appropriate after business conditions changed.

Central research gap. Within the reviewed evidence, there is no jointly evaluated, Bangladesh-oriented retail SME framework that connects traceable operational records to leakage-controlled and calibrated multi-task analytics,

then converts those outputs into constraint-aware, owner-understandable actions whose use and outcomes can be monitored over time.

The central gap is integrative and evaluative. It does not require inventing a new classifier; it requires showing that established methods can be combined without breaking data lineage, temporal validity, decision feasibility or human accountability. Table 2.4 makes each gap falsifiable by linking it to a proposed response and evaluation.

TABLE 2.4: Traceability from identified gap to proposed response and test.

Gap	Proposed response	Evidence required to close the gap	Study link
G1: Context	Tier-1 Bangladesh SME study with explicit external-data boundaries	Primary local records and firm descriptions; no transfer of foreign/generated claims	RQ1–RQ4
G2: Integration	Canonical events, manifests and versioned historical snapshots	Reproducible trace from source row to feature, output, action and later outcome	O1, O2
G3: Validity	Temporal/entity-safe splits, baseline ladder, calibration and frozen test	Out-of-time metrics, ablations, intervals and leakage audit	RQ1, RQ2
G4: Decision	Constraint and expected-utility layer for reorder/retention actions	Same-path policy comparison and sensitivity to cost assumptions	RQ3
G5: Explanation	Selective Bangla evidence with source and uncertainty	Counterbalanced owner tasks measuring correctness, time and confidence	RQ4
G6: Life-cycle	Version, drift, response and outcome monitoring protocol	Revalidation log and post-decision observations	O6

B-SMART is positioned as the proposed response to this combined gap. Public or generated evidence can test parts of the procedure; closing the contextual, decision and owner-centred gaps requires consenting SME data and participant evaluation. If those data are unavailable, the final thesis must report the corresponding gaps as still open rather than treating a prototype or external benchmark as proof of local effectiveness.

Chapter 3

Proposed Methodology

3.1 Research Approach

The thesis will use design science with empirical evaluation. Design science is appropriate because the research produces an artifact intended to address a practical organizational problem and evaluates that artifact against explicit requirements [10]. The artifact is the B-SMART framework, including its canonical data design, time-aware analytical pipeline, constraint-aware recommendation procedure and owner-facing evidence. Empirical study is required because a plausible architecture alone cannot establish predictive validity, usability or business value.

The study is organized into four connected phases:

- P1. **Problem and evidence definition:** establish the target SME setting, research questions, literature-derived requirements, permitted claims and success measures.
- P2. **Artifact design:** specify the operational schema, historical snapshots, candidate analytical methods, strategy constraints, interfaces and audit records.
- P3. **Technical evaluation:** test data integrity, prediction, calibration, robustness and scalability using strictly separated development, public-real and primary-real evidence.
- P4. **Decision and user evaluation:** assess simulated or pilot inventory consequences and test whether owners can understand and correctly use the recommendations.

The unit of forecasting is a branch–SKU–day or branch–SKU–week, selected according to data density. The unit of customer-risk modelling is a customer’s feature snapshot at a historical cutoff. The unit of strategy evaluation is an eligible action evaluated under the same information and constraints as its comparison policy.

3.2 Design Requirements and Acceptance Logic

The literature and problem formulation are converted into eight requirements. They are stated before implementation so that the artifact can be evaluated against a stable purpose rather than judged from whichever functions are easiest to demonstrate.

DR1: Decision relevance.

Every analytical output must name its user, decision, unit, horizon and feasible response. An output that has no defined decision remains exploratory and cannot be promoted as a recommendation.

DR2: Operational integrity.

Source events must retain organization scope, stable keys, event time, receipt time, status and correction history. KPI and model views must be reproducible from those events.

DR3: Temporal validity.

Features, preprocessing parameters and model selection must use only evidence available by the relevant cutoff. Final evaluation must be performed once on a period or entity group not used for selection [15].

DR4: Proportionate modelling.

Every task begins with a transparent baseline. A more complex candidate is accepted only when the available data support it and its improvement is material, stable and relevant to the decision.

DR5: Uncertainty and reliability.

Forecast intervals, probability calibration, data-quality status and sensitivity to uncertain costs must be exposed at the point where they can change an action [19].

DR6: Feasible and responsible action.

Strategy generation must apply cash, stock, lead-time, order-size, contact-consent and policy constraints before ranking. Infeasible actions must be excluded rather than displayed with a low score.

DR7: Human authority and comprehension.

Recommendations must include a concise reason, source and alternative; the owner must be able to inspect, reject or defer the action. Explanation quality is assessed through decision tasks, not inferred from the presence of an attribution [16, 18].

DR8: Accountability and reproducibility.

Every released output must retain the dataset, snapshot, feature, model or rule, constraint and interface versions together with the owner response and any later observable outcome.

Acceptance is conjunctive rather than compensatory. High predictive accuracy cannot compensate for leakage, privacy failure or an infeasible action. A usable interface cannot

compensate for an unreconciled KPI. Conversely, a candidate need not be the most complex model to satisfy the artifact requirements. A simple baseline can be the accepted component when it is temporally valid, adequate for the decision, more stable and easier to maintain.

The evaluation will classify each requirement as *met*, *partially met*, *not met* or *not evaluable with the available evidence*. The last category is important: absence of suitable cost, consent or follow-up data must narrow the conclusion rather than be converted into an assumed success. Requirement-level findings will be reported alongside task metrics so that a single aggregate score cannot hide a critical failure.

3.3 Research Questions, Hypotheses and Measures

The research questions introduced in Chapter 1 are operationalized as follows:

RQ1 / H1.

A demand model containing local calendar, price, promotion and available market-context features will reduce out-of-time WAPE or MASE relative to the otherwise identical model without those feature groups and relative to a seasonal-naïve baseline.

RQ2 / H2.

A model trained from historical customer snapshots will identify future inactivity better than the event base rate, measured by PR-AUC, and will achieve Lift@10% above 1.0 while retaining acceptable probability calibration.

RQ3 / H3.

Under identical demand sequences, costs and constraints, the proposed forecast-driven replenishment policy will reduce total stock-out and holding cost relative to the participating owner's documented current policy.

RQ4 / H4.

In counterbalanced decision tasks, an explanation condition will improve task correctness or confidence without a practically harmful increase in completion time compared with an output-only condition.

TABLE 3.1: Alignment of research questions, evidence and primary measures.

RQ	Unit and comparison	Primary measures	Required evidence
RQ1	Unseen branch-SKU periods; context model versus ablations/baselines	WAPE, MASE, interval coverage	Time-stamped sales, stock, price, promotion and calendar data
RQ2	Future customer windows; model versus base-rate/rule baseline	PR-AUC, Lift@10%, Brier score, calibration	Pseudonymous customer transactions with historical cutoffs
RQ3	Same demand path; proposed versus current replenishment policy	Stock-out units, holding units, total cost	Stock, lead time, order constraints and validated cost assumptions
RQ4	Same decision scenarios; explanation versus output only	Correctness, time, confidence, perceived usability	Consenting target users and counterbalanced tasks

Hypotheses and primary measures are fixed before the final test is inspected. Secondary analyses will be labelled as exploratory. A statistically detectable difference will not be described as practically important unless its magnitude or cost implication is material.

3.4 Target Population and Sampling Plan

The target population comprises inventory-based retail SMEs operating in Bangladesh. To reduce uncontrolled variation, initial recruitment will focus on three to five businesses from one broadly comparable retail vertical. The planned analytical target is 12–24 months of records across the partners, subject to ethics approval and data availability. A first partner with a shorter history may be used only to test collection, mapping and data-quality rules.

Businesses will be recruited purposively because access to longitudinal invoice and stock data requires owner agreement. Inclusion requires identifiable dates, products and quantities and a person able to explain the current replenishment process. Customer-level analysis additionally requires stable pseudonymous customer identifiers and sufficient follow-up after each cutoff. A business or task will be excluded from a particular analysis when these minimum conditions are absent; it will not be repaired by inventing identifiers or outcomes.

3.5 Data Sources and Evidence Tiers

Evidence will be kept in separate tiers so that convenience data do not silently change the claim being tested.

TABLE 3.2: Planned evidence tiers and claim boundaries.

Tier	Source	Permitted use	Prohibited inference
1	Consenting Bangladesh SME invoice, stock and operating records	Primary local forecasting, customer, strategy and usability evaluation	Generalization beyond sampled firms without further evidence
2a	Bangladesh WFP/DAM market-price data [25]	Local external context and separate price-series study	SKU sales, customer behavior or firm revenue
2b	UCI Online Retail II, United Kingdom [3]	Real-invoice pipeline and method portability checks	Bangladesh purchasing behavior
2c	Public Bangladesh retailer demand series [14]	Local quantity-forecast procedure check	Multi-SKU, customer, price or profit claims if fields are absent
3	Clearly labelled generated records	Software tests, rare cases, scale tests and privacy-safe examples	Field effectiveness or real customer behavior

Every received dataset will have a manifest containing source, owner or repository, collection period, receipt date, checksum, license or agreement, row count, schema version, mapping, exclusions and known limitations. Tier-3 data will never be merged into a Tier-1 test set. When augmentation is examined, the real-only and augmented training conditions will be reported separately and evaluated on the same untouched real test period.

3.6 Data Acquisition and Variable Plan

Primary collection will prefer exports from the business’s existing point of sale or spreadsheet workflow. A controlled template will be offered when no consistent export exists. The minimum sales fields are timestamp, business, branch, invoice, SKU, quantity, unit price, discount and cancellation/return status. Recommended fields include unit cost, promotion, payment mode, pseudonymous customer, stock-on-hand, stock movement, purchase receipt, supplier lead time and incoming quantity.

Variables are organized by the decision time at which they become available:

TABLE 3.3: Feature availability and leakage rule.

Availability class	Examples	Rule
Known before cutoff	Historical sales, current stock, recorded price, lead time, calendar	Eligible if timestamp and provenance are valid
Known for forecast horizon	Confirmed promotion, planned closure, calendar event	Eligible only when the same information would exist in real use
Observed after outcome	Later purchase, delivered satisfaction, realized return, future stock adjustment	Label or evaluation outcome only; never a predictor
Uncertain or revised	Unconfirmed promotion, estimated cost, manual lead time	Retain source and uncertainty; include a sensitivity analysis

3.7 Data-Quality Assessment and Task Eligibility

Quality assessment will be performed before feature engineering and repeated after every structural transformation. A dataset-wide average will not be used as the sole quality claim because it can conceal failure in a field essential to one task. Instead, a quality profile will be calculated by business, branch, field, period and analytical task.

For required field j in eligible row set G , completeness is

$$C_{j,G} = 1 - \frac{\sum_{i \in G} \mathbb{I}(x_{ij} \text{ is missing})}{|G|}. \quad (3.1)$$

Completeness does not imply validity. Product codes will be tested for referential integrity; invoice-line totals will be recomputed from quantity, price and discount; returns and cancellations will be reconciled to their source order when the identifier exists; and payment totals will be compared with invoice settlement status. Duplicate detection will distinguish exact repeated rows from legitimate repeated purchases. Candidate duplicates will be logged and resolved by a declared rule rather than deleted solely because their values are similar.

Inventory provides a structural reconciliation test. Let $I_{p,t}^{\text{open}}$ and $I_{p,t}^{\text{close}}$ denote the recorded opening and closing balance for product p over interval t , and let $\Delta I_{p,t,m}$ be a signed movement such as receipt, sale, return, damage or manual correction. The residual

$$r_{p,t} = I_{p,t}^{\text{close}} - I_{p,t}^{\text{open}} - \sum_m \Delta I_{p,t,m} \quad (3.2)$$

should be zero when all movements and balances are recorded consistently. A nonzero residual is not silently forced to zero. Its magnitude, period and likely source are retained as a quality event, and affected observations are flagged for sensitivity analysis or exclusion.

Temporal tests will identify events recorded before their source transaction, receipts before purchase orders, negative or implausible lead times, customer features after their cutoff and product histories outside the active period. Event time and ingestion time will be stored separately where available. When only a date is known, the system will not invent false within-day precision.

Task eligibility is then determined from necessary evidence rather than from a single arbitrary missing-data threshold. Demand forecasting requires a stable product-time series and a defensible availability interpretation. Margin KPIs require cost provenance. Customer inactivity requires stable pseudonymous identity and sufficient future follow-up. Replenishment evaluation additionally requires stock or a declared inventory reconstruction, lead-time assumptions and cost ranges. Owner-facing retention requires consent and a permitted contact mechanism. A task-business-period combination that fails a necessary condition is labelled ineligible; the reason becomes part of the audit report.

This gate prevents statistical imputation from creating evidence that the business never recorded. Imputation remains permissible for a model feature when its missingness mechanism and decision-time availability are understood, the transformer is fitted on training data only, and an ablation or sensitivity check shows whether the choice materially affects results. It is not permissible to impute identifiers, consent, future outcomes or the ground truth used for the primary test.

3.8 Ethics, Privacy and Governance

University ethics approval and a signed data-sharing agreement will precede primary collection. Direct identifiers such as names, phone numbers, e-mail addresses, national identifiers, account numbers and wallet numbers are not required for the research tables. Each business will use a separate salted pseudonymous customer key; any re-identification mapping will remain with the business and outside the research repository.

The agreement will state research purpose, authorized researchers, storage, retention deadline, publication form, withdrawal and deletion procedure. Access will follow least privilege, and published material will use aggregates or separately reviewed de-identified examples. A retention suggestion will not authorize contact: customer consent and the participating business's policy are independent feasibility constraints. Model outputs will support, not replace, owner decisions.

3.9 Canonical Operational and Analytical Model

The proposed operational schema contains Organization, Branch, Product, Customer, Supplier, SalesOrder, SalesOrderItem, Payment, InventoryBalance, StockMovement, PurchaseOrder, Expense, LedgerEntry, SalesReturn, Refund, ImportBatch and AuditLog entities. The analytical layer does not train directly on mutable operational tables. It creates versioned invoice-line, SKU-time and customer-cutoff views with the source keys and data cutoff retained.

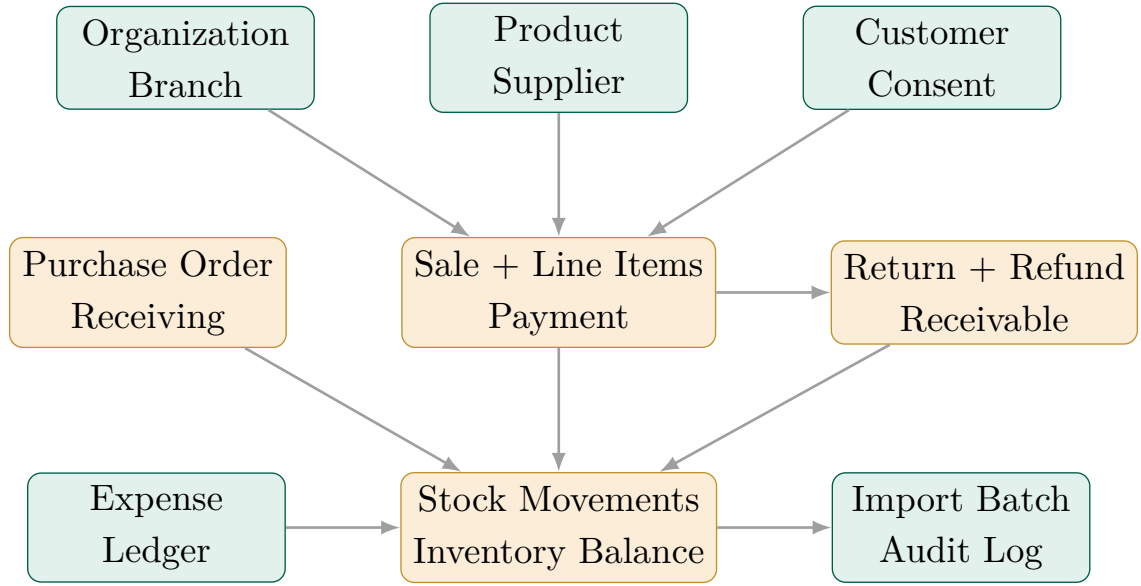


FIGURE 3.1: Proposed canonical operational data model.

Validation will test uniqueness of business-scoped identifiers, chronological consistency, nonnegative physical quantities where appropriate, payment/order reconciliation, cancellation handling and stock-movement balance. Corrections will be recorded as transformations rather than silently overwriting the source.

3.10 Formal Problem Formulation

For business b , branch l , product p , customer c and time t , let

$$D_t = \{\text{Sales, Stock, Purchases, Expenses, Customers, Payments, Context}\}_{\leq t}. \quad (3.3)$$

3.10.1 Performance Evaluation

For each KPI K , the system returns $K(D_t; w, g)$, where w is the reporting window and g is the aggregation level. Each value retains numerator, denominator, filters and data-quality status so that the owner can reproduce it.

For valid invoice lines i in reporting window W , net sales will be defined from the business's documented discount, cancellation and return treatment. A general form is

$$\text{NetSales}_W = \sum_{i \in W} q_i p_i - \sum_{i \in W} d_i - \sum_{r \in W} v_r, \quad (3.4)$$

where $q_i p_i$ is gross line value, d_i is the recognized discount and v_r is the value of valid returns or reversals assigned to the window. Gross margin is reported only when comparable cost evidence exists:

$$\text{GrossMarginRate}_W = \frac{\text{NetSales}_W - \text{COGS}_W}{\text{NetSales}_W}. \quad (3.5)$$

If cost is missing for a material share of sold units, the system reports sales and cost coverage rather than silently interpreting revenue as profit. Average order value uses eligible completed orders as the denominator; repeat rate states the customer-identification coverage; stock turnover states both the cost and average-inventory convention. Period comparisons will use comparable windows or explicitly label trading-day and closure differences.

Each KPI response consists of a value, definition version, period, aggregation, coverage indicators and reconciliation status. This structure makes a change in business-rule definition visible. It also permits the evaluation to separate a calculation defect from a genuine change in business performance.

3.10.2 Demand Prediction

$$\hat{y}_{b,l,p,t+h} = f\left(X_t^{\text{history}}, X_t^{\text{price}}, X_t^{\text{promotion}}, X_{t:t+h}^{\text{calendar}}, X_t^{\text{market}}\right), \quad (3.6)$$

where $\hat{y}$ is expected demand at horizon h . Quantile forecasts or calibrated residual intervals will represent uncertainty. When product-level forecasts are aggregated, reconciliation will be considered so SKU, category, branch and business totals remain coherent [12].

3.10.3 Future Customer Inactivity

$$q_{c,t} = P(\text{no purchase in } (t, t + H] \mid X_{c,t}), \quad (3.7)$$

where $X_{c,t}$ contains only records available by cutoff t . Horizon H will be fixed from business purchase cycles before final evaluation.

3.10.4 Strategy Optimization

For an eligible action $a \in A_t$,

$$U(a \mid D_t) = E[\Delta \text{Profit}(a)] - \text{ActionCost}(a) - \lambda \text{Risk}(a), \quad (3.8)$$

subject to budget, storage, minimum order quantity, supplier lead time, current/incoming stock and contact consent. This is a transparent operational objective inspired by the predictive-to-prescriptive transition [1]; its terms will be estimated and sensitivity tested rather than presented as known constants.

For replenishment, let the inventory position immediately before action be

$$\text{IP}_{p,t} = \text{OnHand}_{p,t} + \text{OnOrder}_{p,t} - \text{Backorder}_{p,t}. \quad (3.9)$$

Let $D_{p,t}^{(L)}$ be demand over the uncertain supplier lead-time horizon and Q a candidate order quantity. A transparent conditional cost is

$$J_p(Q) = c_o \mathbb{I}(Q > 0) c_h E[(\text{IP}_{p,t} + Q - D_{p,t}^{(L)})^+] c_s E[(D_{p,t}^{(L)} - \text{IP}_{p,t} - Q)^+], \quad (3.10)$$

where c_o , c_h and c_s denote declared ordering, holding and shortage costs. The feasible set enforces supplier pack size, minimum order, storage, available cash and any category-level purchasing limit. Equation 3.10 is an evaluation model, not a claim that the three costs are known exactly. Candidate policies will be compared across owner-reviewed ranges, and the system will withhold a precise savings claim when the ranking changes materially across plausible values.

Uncertainty enters through the lead-time demand distribution, not through a single inflated point forecast. When reliable quantiles are available, the system can evaluate candidate quantities against those quantiles; otherwise it uses residual or replay-based scenarios from validation periods. If stock-out censoring prevents credible demand reconstruction, the result is reported as a policy comparison under observed sales rather than as an estimate of fully unconstrained demand.

Customer action requires a different logic. The inactivity probability $q_{c,t}$ in Equation 3.7 can prioritize review, but it does not estimate the causal effect of contacting customer c . A contact action is eligible only if the customer is contactable under consent and business policy, the proposed treatment is defined and capacity remains. Expected profit will be calculated only if a defensible incremental response benefit is available; otherwise

the output is explicitly a risk-ranked contact list. This separation prevents a predictive probability from being misrepresented as campaign uplift.

The action layer can also decline to recommend. Abstention occurs when required constraints are missing, uncertainty makes the top alternatives practically indistinguishable, the predicted benefit does not exceed action cost over the declared sensitivity range, or the relevant model has failed a monitoring gate. An abstention with a reason is treated as a valid safety behavior rather than a system error.

3.11 Preprocessing and Historical Snapshots

The processing order is fixed: validate source and schema; remove or mark exact duplicates; resolve cancellations and returns; construct the split; fit imputation, encoding and scaling on training data; build features; train; and evaluate once on the untouched test. The order prevents statistics from future periods entering training [15].

Demand features may include lags, rolling location and variability, day of week, month, festival distance, price, discount, known promotion, stock-out indicator and external market context. Zero demand caused by no sale is distinguished, when possible, from zero observed sales caused by no stock. Intermittency will be measured before choosing temporal granularity and candidate methods.

Customer snapshots will include RFM, tenure, order-value statistics, category breadth, purchase interval, discount behavior and pre-cutoff return rate. The label window follows the observation window, with a purge interval when feature construction or delayed events could overlap it. No customer identifier appears on both sides of an entity-generalization test.

3.12 Candidate Methods and Selection Rules

Method selection follows a baseline ladder, and eligibility depends on evidence volume and task structure.

- **Performance:** auditable KPI definitions and rule-based anomaly flags establish the descriptive baseline.
- **Demand:** last-value, seasonal-naive, moving-average or exponential-smoothing baselines precede ARIMA/SARIMA, Croston-family methods, gradient-boosted lag models and, only with adequate sequences, LSTM. Croston and the Syntetos–Boylan correction address intermittent demand [6, 24].
- **Inactivity:** a recency or base-rate rule is compared with Logistic Regression, Random Forest [2] and XGBoost [5]. Class weighting and decision threshold are selected on development data.

- **Segmentation:** scaled RFM features are studied with K-Means. Candidate K values are judged by inertia, silhouette, resampling stability and business interpretability.
- **Explanation:** coefficient/rule explanations are preferred for simple models; SHAP supplies model-behavior attributions for tree models [16]. Displayed reasons will remain selective and contextual [18].

A complex candidate will be rejected when it lacks sufficient training history, cannot beat the relevant baseline on development data, produces unstable probabilities, or adds operational cost without material value. This rule keeps the proposed system useful when the scientifically honest answer is a simple forecast or business rule.

3.13 The B-SMART Sequential Procedure

Algorithm 1: B-SMART Strategy Recommendation

Input: Data available at cutoff D_t , horizon H , constraints C_t , validated model/baseline versions M , candidate actions A

Output: Ranked feasible recommendations R_t

- 1 Verify source, schema, timestamps and organization ownership;
 - 2 Create the historical feature snapshot X_t ;
 - 3 Apply transformations fitted on the appropriate training evidence;
 - 4 Estimate demand with uncertainty and customer inactivity probabilities;
 - 5 Calculate reproducible KPIs and data-quality warnings;
 - 6 Generate candidate actions with source and decision-time metadata;
 - 7 **foreach** $a \in A$ **do**
 - 8 Check feasibility under C_t ;
 - 9 Estimate benefit, cost and uncertainty using declared assumptions;
 - 10 $U(a) \leftarrow E[\Delta\text{Profit}(a)] - \text{Cost}(a) - \lambda\text{Risk}(a)$;
 - 11 Remove infeasible actions and rank the remainder by $U(a)$;
 - 12 Attach concise explanation, uncertainty, cutoff and evidence source;
 - 13 Return top- K actions; record owner response and later observable outcome;
-

For N invoice lines, d validated fields and P candidate actions, source validation and feature aggregation are approximately $O(Nd)$ and action ranking is $O(P \log P)$. Training complexity depends on the selected model and is measured empirically. Serving reuses versioned artifacts; it must not retrain in the request path.

3.14 Proposed Architecture and Information Flow

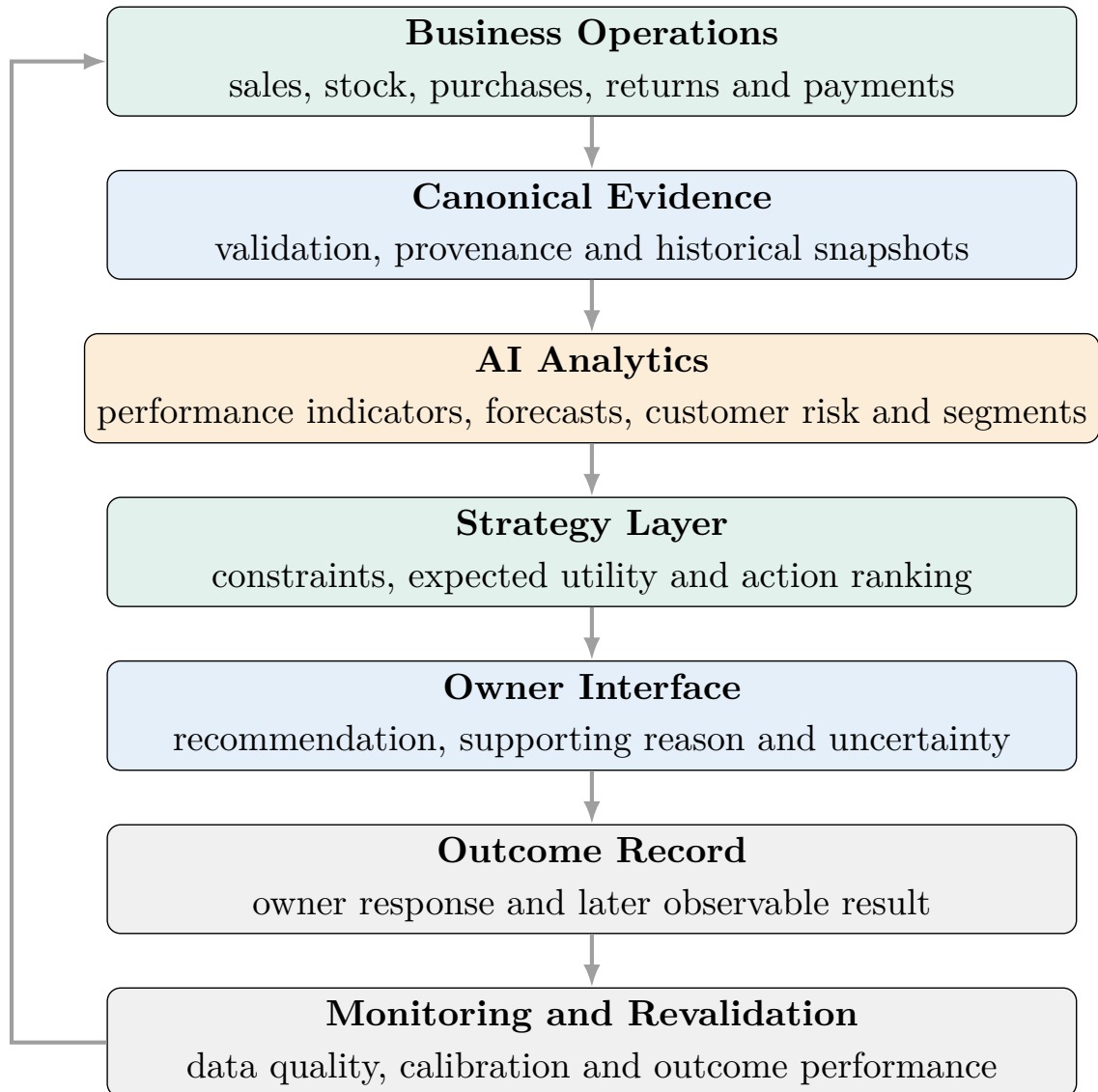


FIGURE 3.2: Proposed B-SMART architecture and accountable feedback loop.

The information flow is intentionally one-directional at prediction time: future outcomes cannot alter the snapshot that produced an earlier result. Outcomes enter later through the monitoring path and become eligible training evidence only in a new, versioned cycle. This separation makes it possible to reproduce what the system knew when an action was proposed.

3.15 Snapshot Lineage and Frozen Evaluation Flow

The analytical snapshot is the unit that prevents future information from entering an earlier decision. Figure 3.3 distinguishes the observation window, decision cutoff and outcome window. Raw events remain in the governed ledger. A snapshot builder selects records available by cutoff t , applies a declared mapping and produces task-specific features. The snapshot identifier, cutoff, source-manifest hash and feature-definition version are immutable.

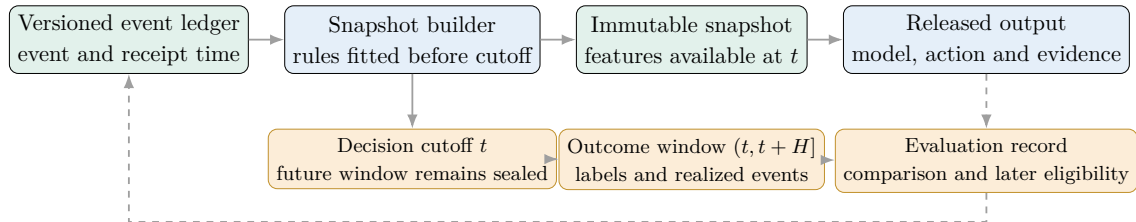


FIGURE 3.3: Historical snapshot construction and separation of later outcomes.

For forecasting, multiple cutoffs form rolling origins. Transformations such as scalers, encoders, imputers and feature-selection rules are fitted within each training window and then applied to its validation origin. For customer inactivity, each snapshot contains behavior up to t and receives a label only after the full horizon H is observable. A purge interval is inserted when a delayed return, fulfillment event or rolling feature could cross the boundary. For entity-generalization analysis, the customer or product grouping is applied before any learned transformation.

Model development uses training and validation snapshots. The final test manifest remains sealed until the task definition, eligibility rules, feature set, candidate models, primary metrics, thresholds and random seeds have been frozen. If a defect is discovered after opening the test, the correction and its impact are documented; the same observations cannot be repeatedly reused as if they were still an untouched test. This rule protects the interpretation of the final result rather than merely the mechanics of a split.

Outcome records do not flow back into the released model automatically. They first enter a monitoring store and become eligible for a later version only after reconciliation, label maturation and repetition of the validation protocol. This delay prevents an owner response, corrected invoice or partially observed horizon from changing the evidence behind a historical output.

3.16 Owner-Facing Recommendation Evidence Contract

Every displayed recommendation will conform to a small evidence contract. The contract separates the suggested action from the information needed to judge it, as illustrated in Figure 3.4. The interface may show fewer details initially, but the full record must remain available for audit and evaluation.

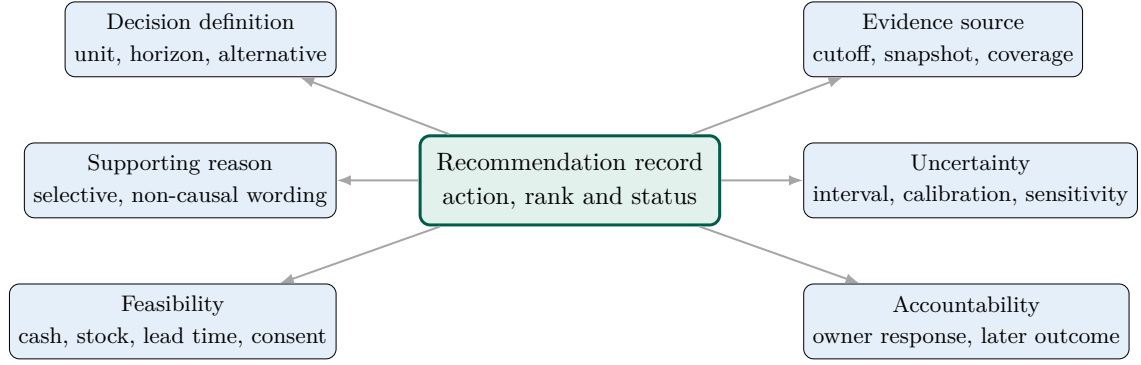


FIGURE 3.4: Minimum evidence contract for an owner-facing recommendation.

The action component states exactly what is proposed and what comparator would otherwise be used. The evidence component provides the cutoff, source and coverage so the owner can detect stale or incomplete data. The reason component uses a small number of faithful factors and avoids causal language. Uncertainty and sensitivity state whether nearby alternatives are practically different. The feasibility component records every binding constraint. Finally, the accountability component captures accept, reject, defer or modify together with an optional owner reason and the later observable outcome.

An explanation is considered successful only when it supports correct use. The owner study will therefore include cases in which the right decision is to reject or defer a confident-looking recommendation. This tests calibration of human reliance, not only whether participants prefer a more detailed screen. The contract also supports failure analysis: a poor decision can be traced to the underlying estimate, missing constraint, explanation, stale data or owner interpretation instead of being attributed vaguely to “the AI.”

3.17 Evaluation Protocol

3.17.1 Forecasting Evaluation

All forecasting splits preserve time. Candidate methods are selected through rolling-origin or expanding-window validation, followed by one evaluation on a final untouched period. For actual values y_i and forecasts $\hat{y}_i$,

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (3.11)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (3.12)$$

$$\text{WAPE} = \frac{\sum_i |y_i - \hat{y}_i|}{\sum_i |y_i|}. \quad (3.13)$$

MASE will support comparison across scales and avoid some defects of percentage errors [13]. Prediction-interval coverage and mean interval width will evaluate uncertainty. Results will be stratified by demand volume and intermittency rather than hiding failure on low-volume items in one aggregate.

3.17.2 Customer and Segmentation Evaluation

Inactivity models will report outcome prevalence, PR-AUC, ROC-AUC, Lift@10%, Brier score and reliability plots. The emphasis on PR measures follows their value under imbalance [21]; calibration is assessed separately because a useful ranking can still produce misleading probabilities [19]. Thresholds and contact fractions are chosen on validation evidence using declared costs, not adjusted after viewing the test.

Clustering evaluation will report inertia, silhouette, cluster stability, minimum cluster size and RFM profiles. The final interpretation will be reviewed with domain participants. A low-separation result will be reported as such; the existence of clusters will not be treated as proof of natural customer types.

3.17.3 Ablation, Robustness and Statistical Analysis

Planned ablations remove one feature family at a time: local calendar, price/promotion, market context and customer-history extensions. Additional comparisons examine global versus business-specific models and real-only versus clearly controlled augmentation. All paired comparisons use the same test instances or forecast origins.

Uncertainty will be summarized with paired block bootstrap intervals across time or business units, as appropriate. Model differences will be reported with effect size and interval, not only a significance label. Sensitivity analysis will vary lead time, shortage cost, holding cost, action cost and risk penalty in Equation 3.8.

3.17.4 Frozen Analysis and Statistical Decision Rules

For each research question, one primary comparison and its primary metric will be frozen before the final test is opened. RQ1 uses paired out-of-time forecast errors at common origins, with WAPE or MASE selected according to denominator and scale suitability. RQ2 uses PR-AUC as the principal discrimination measure and treats calibration as a separate acceptance condition. RQ3 compares policy cost on identical demand paths under the same opening inventory and constraints. RQ4 compares task correctness between explanation conditions, with completion time and confidence interpreted jointly rather than as independent proof of benefit.

The unit of uncertainty will match the dependence structure. Forecast errors from adjacent dates are not treated as independent rows; block bootstrap resampling will preserve

local temporal dependence. When multiple businesses are available, firm-level variation will be shown rather than hidden inside a pooled transaction count. Customer comparisons will respect repeated snapshots from the same customer, and usability tasks will be paired within participant when the design permits. Confidence intervals will accompany effect estimates so that imprecision remains visible.

Practical importance will be declared separately from statistical evidence. A small average error reduction may be practically relevant for a high-volume item but irrelevant after order constraints; a change in task time may be acceptable if correctness improves materially. Cost or effect thresholds will be defined from the participating business context before final comparison. Where no defensible threshold exists, the report will present the magnitude and uncertainty without assigning an arbitrary success label.

Secondary metrics, horizons, segments and feature interactions remain useful for diagnosis, but they will be labelled secondary or exploratory. If many related comparisons are inspected, the report will disclose their number and avoid selecting only favorable cases. Any post-test model or threshold change creates a new exploratory version; it does not replace the frozen primary result. This separation keeps model improvement possible without rewriting the original evidence.

Missing outcomes will be counted and explained. An observation with an incomplete label horizon cannot be a negative inactivity case. An unknown stock balance cannot be treated automatically as zero stock. A recommendation without later outcome may still support usability analysis but cannot support realized impact. Sensitivity bounds will be used when assumptions can be varied plausibly; otherwise the corresponding claim will be marked not evaluable.

3.17.5 Decision, Usability and Scalability Evaluation

Replenishment policies will face identical observed or replayed demand paths. The comparison records stock-out units, average inventory, holding cost, shortage cost and total cost. Retention ranking will use expected value only when response benefit, contact cost and consent are available; otherwise it will remain a prioritization study, not a profit claim.

Owner evaluation will use realistic tasks in a counterbalanced within-subject design when the sample permits. Measures include task correctness, completion time, error count, confidence and perceived usability. Participants may explain their decision in their own words, allowing the study to detect plausible but misunderstood explanations.

Scalability tests will use generated data only and will remain separate from accuracy results. Planned sizes are 100, 500, 1,000, 10,000, 100,000 and 1,000,000 transaction lines. Measures include import and preprocessing time, training time, prediction latency, API throughput, peak memory, artifact size and representative query latency. Any association-rule support experiment will be reported only after the corresponding algorithm exists and has been run.

3.17.6 Monitoring Plan

Monitoring will distinguish schema or data-quality failure, input shift, calibration change and outcome-performance change. Alerts will not automatically authorize retraining. A new model must repeat the validation protocol and retain its training window, features, parameters and comparison result. This follows the methodology's requirement that adaptation must itself be evaluated before deployment.

3.18 Reproducibility, Audit and Failure Reporting

Reproducibility will be designed at four levels. *Data reproducibility* requires a source manifest, checksum, schema version, mapping and exclusion log. *Analytical reproducibility* requires snapshot identifiers, fitted transformations, feature definitions, target construction and split manifests. *Model reproducibility* requires code version, environment, parameters, random seeds, trained artifact and evaluation command. *Decision reproducibility* additionally requires constraints, cost assumptions, candidate-action set, ranking rule and interface version.

Primary partner data cannot be published merely to make the study reproducible. The reproducibility package will therefore separate restricted evidence from shareable artifacts. It can publish schemas, data dictionaries, manifest templates, transformation code, generated test fixtures, configuration files and commands without releasing identifiers or confidential transactions. A permitted evaluator with the restricted dataset should be able to regenerate the corresponding snapshot and result; an external reader should be able to inspect the procedure and run the software on a documented public or generated fixture.

Every experiment will create a machine-readable run record containing the research question, dataset and snapshot hashes, training and test windows, grouping keys, feature version, candidate and baseline versions, parameters, metrics, warnings, start and finish times, and output artifact hashes. Manual spreadsheet corrections will not occur outside this record. If an owner supplies a correction, it enters as a dated source or mapping revision and triggers only the affected downstream views.

Failure reporting is part of the scientific output. Data eligibility failure, training failure, numerical instability, non-convergence, resource exhaustion, constraint infeasibility and evaluation failure will have distinct statuses. The report will give the denominator of attempted cases and will not compute an average only over successful model runs without disclosure. A method that fails on sparse or short series is therefore compared as deployed, not only on the subset where it happens to return a prediction.

Auditability also applies to the owner interface. Each recommendation receives an identifier linking its evidence contract, displayed text, owner response and later outcome. If a recommendation changes after data correction or model replacement, the earlier record

remains immutable and the successor points to it. This supports accountability without implying that the model, researcher or owner alone caused the eventual outcome.

The final research package will include a claim-to-evidence index. Each central claim will identify the relevant research question, figure or table, run record and evidence tier. Claims supported only by public, generated, simulated or small-sample evidence will remain labelled accordingly. This index provides a final check that polished presentation has not expanded the scope of what was actually evaluated.

3.19 Threats to Validity and Planned Controls

TABLE 3.4: Principal validity threats and planned controls.

Threat	How it may distort evidence	Planned control
Selection bias	Cooperative firms may be more digitally mature	Describe recruitment and firms; restrict generalization
Leakage	Future or target-derived data inflate performance	Availability table, cutoff snapshots, grouped/temporal split
Stock-out censoring	Observed sales understate unconstrained demand	Retain stock state; stratify or flag censored periods
Small/short series	Complex models overfit	Eligibility rule, simple baselines, rolling validation
Dataset mismatch	Foreign/public tasks are mistaken for local evidence	Evidence tiers and separate result tables
Uncertain costs	Utility ranking depends on weak assumptions	Owner-validated ranges and sensitivity analysis
Novelty/learning effect	Interface condition benefits from presentation order	Counterbalance tasks and collect task-level timing
Concept drift	Festival, price or behavior relations change	Time-based tests and post-deployment monitoring plan

3.20 Expected Methodological Outputs

At completion, the methodology is expected to yield a documented canonical schema and data dictionary, partner data protocol, versioned feature snapshots, baseline and candidate model specifications, a constraint-aware recommendation procedure, a reproducible evaluation package and an owner-study instrument. Implementation detail, observed numerical results and conclusions will be added to the reserved chapters only after the corresponding evidence is available.

Chapter 4

Implementation

Chapter 5

Experimental Results

Chapter 6

Conclusions

6.1 Current-Stage Conclusion

This initial report establishes a focused foundation for an AI-powered business analytics system for Bangladeshi retail SMEs. The central problem is not merely the absence of prediction models. It is the missing, trustworthy connection between operational records, temporally valid analytics and a feasible owner decision. A forecast has limited practical value if its source cannot be traced, its test contains future information, or the result ignores cash, stock and lead time. A customer-risk score is similarly incomplete when its probability is uncalibrated or the proposed contact does not respect consent.

The evidence leads to four conclusions. Bangladesh-specific studies make workflow fit, perceived value, cost and local technology use necessary design considerations [9, 11, 22]. Retail and inventory research requires explicit treatment of intermittency, hierarchy, forecast error and stock consequences [8, 12, 23, 24]. Leakage control, imbalance-aware evaluation, probability calibration and understandable evidence are scientific safeguards [15, 16, 19, 21]. The resulting research gap is integrative: these principles have not yet been evaluated together in a bounded Bangladesh-oriented SME workflow.

The proposed B-SMART framework responds through a traceable sequence of operational events, canonical evidence, historical snapshots, baseline/model comparison, uncertainty and explanation, constraint-aware action ranking, owner response and later outcome monitoring. Its proposed contribution is this governed sequence and its evaluation, not the invention of a new classifier or an unsupported claim that a complex model will always perform best.

The report therefore supplies a testable path into the next phase: ethics and partner-data readiness, model freeze, untouched evaluation, usability tasks and decision-impact analysis. It does not yet establish system effectiveness. Final claims must follow the observed evidence, including any result in which a simple baseline wins, explanations show no measurable advantage, or the available data cannot support a research question.

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